

AGROCLIMATIC ZONES

Agro-climatic zones is a land **unit area uniform in respect of climate and length of growing** period (LGP) which is climatically suitable for a certain range of crops and cultivars (FAO, 1983).

Classification by Planning Commission of India (1989) made an attempt to delineate the country into different Agro climatic regions based on homogeneity in rainfall, temperature, topography, cropping and farming systems and water resources.

India Is Divided Into 15 Agro-Climatic Regions

1. Western Himalayan zone-

This zone consists of three **distinct sub-zones of Jammu and Kashmir, Himachal Pradesh and Uttar Pradesh hills.**

Features

- ✓ The region **consists of skeletal soils of cold region, podzolic** mountain meadow soils and hilly brown soils.
- ✓ Lands of the region **have steep slopes in** undulating terrain.
- ✓ Soils are **generally silty loams** and these are prone to erosion hazards.

Crop cultivation- plantation crop like almond, walnut, apple, apricots etc cereals- wheat

Animal domestication -goat, yak, sheep etc

2. Eastern Himalayan zone-

Sikkim and **Darjeeling hills, Arunachal Pradesh, Meghalaya, Nagaland, Manipur, Tripura, Mizoram, Assam** and Jalpaiguri and Cooch Bihar districts of West Bengal fall under this region,

Features

- ✓ **High rainfall and high forest cover.**
- ✓ Shifting cultivation is practiced **in nearly one-third of the cultivated area** and this has caused denudation and degradation of soils.
- ✓ **Heavy runoff, massive soil erosion** and floods in lower reaches and basins.

Major crop cultivation- tea plantation, rice, maize, vegetables etc & shifting cultivation

Animal domestication -pig, sheep, cattle ex **Mithun breed**

3. Lower Gangetic Plains zone-

This zone consists of West Bengal-lower Gangetic plain region.

Feature

are mostly **alluvial and are prone to floods.**

cultivation – rice & jute, vegetables ex cauliflower

domestication-cattle, sheep, fishery etc.

4. Middle Gangetic Plains zone

This zone consists **of 12 districts of eastern Uttar Pradesh and 27 districts** of Bihar plains.

Features

- ✓ This zone has a geographical **area of 16 million hectares** and rainfall is high.
- ✓ About **39% of gross cropped area** is irrigated and the cropping intensity is 142%.

Highly cropped region as all the **major crops like rice, wheat, mustard**, sugarcane, vegetables etc

Animal domestication-cattle, **buffalo, sheep, poultry** etc

5. Upper Gangetic Plains zone –

This zone consists of **32 districts of Uttar Pradesh**. Irrigation is through canals and tube wells.

- ✓ A good potential for **exploitation of ground** water exists

6. Trans-Gangetic Plains zone–

This zone consists of **Punjab, Haryana, Union territories of Delhi and Chandigarh** and Sri Ganganagar district of Rajasthan.

Features

- ✓ **Highest net sown area,**
- ✓ **Highest irrigated area**
- ✓ **High cropping intensity**
- ✓ **High groundwater utilization.**

Crop cultivation- rice, wheat, millets, mustard, sugarcane etc.

Animal domestication-cattle, buffalo, goat, piggery, poultry etc.

7. Eastern Plateau and Hills zone –

This zone consists of **eastern part of Madhya Pradesh, southern part of West Bengal and most of inland Orissa.**

Features

- ✓ The soils are **shallow and medium in depth**
- ✓ The topography is **undulating with a slope of 1-10%.**
- ✓ Irrigation is **through tanks and tube** wells.

Crop cultivation- wheat, ragi, sorghum, potato etc.

Animal domestication-cattle, goat, sheep etc.

8. Central Plateau and Hills zone–

This zone comprises of **46 districts of Madhya Pradesh, part of Uttar Pradesh** and Rajasthan.

Features

- ✓ The topography is **highly variable** nearly **1/3rd of the land is** not available for cultivation.
- ✓ Irrigation and cropping **intensity are low**.
- ✓ 75% of the area is **rainfed** grown with **low value cereal** crops.
- ✓ There is an **intensive need for alternate high value crops including** horticultural crops.

Crop cultivation- cotton, wheat, millets (bajra), jowar, soyabean, pulses (green gram), sesamum etc

Animal domestication- poultry, cattle, goat, sheep etc.

9. Western Plateau and Hills zone-

This zone comprises the major part of **Maharashtra, parts of Madhya Pradesh** and one district of Rajasthan.

Features

- ✓ The average rainfall of **the zone is 904 mm**.
- ✓ The net sown **area is 65% and forests occupy 11%**.
- ✓ The **irrigated area is only 12.4%** with canals being the main source.

Crop cultivation -jowar, cotton, sugarcane, rice, bajra, potato etc.

Animal domestication-cattle, goats, poultry, piggery etc.

10. Southern Plateau and Hills zone-

This zone comprises **35 districts of Andhra Pradesh**, Karnataka and Tamil Nadu which are typically semi-arid zones.

Features

- ✓ Dryland farming **is adopted in 81% of** the area and the **cropping intensity is 111** percent.

Crop cultivation- millets, cotton, rice, ragi wheat, vegetables, spices etc

Animal domestication- draught breeds of cattle, goat, buffalo etc.

11. East Coast Plains and Hills zone-

This zone comprises of **east coast of Tamil Nadu, Andhra Pradesh and Orissa.**

Features

- ✓ Soils are **mainly alluvial and coastal sands.**
- ✓ Irrigation is **through canals and tanks.**

Crop- cultivation-rice along with fishery (Godavari basin), species & condiments.

12. West Coast Plains and Ghats zone-

This zone comprises **west coast of Tamil Nadu, Kerala, Karnataka, Maharashtra** and Goa with a variety of crop patterns, rainfall and soil types

Crop cultivated- cashew, millets, sugarcane, rice, **spices, ginger, turmeric** etc

Animal domesticated- fishery, cattle, poultry, goat etc.

13. Gujarat Plains and Hills zone-

This zone consists of 19 districts of Gujarat.

Features

- ✓ This zone is arid with **low rainfall in most parts.**
- ✓ only **32.5% of the** area is irrigated largely through wells and tube wells.

Crop cultivated – groundnut, cumin, millets etc

Animal domestication- goat, cattle, camel, poultry etc.

14. Western Dry zone-

This zone comprises **nine districts of Rajasthan**

Features

- ✓ Hot sandy **desert, erratic rainfall, high evaporation**, scanty vegetation.
- ✓ The ground **water is deep and often brackish.**
- ✓ **Famine and drought** are common features of the region.

Crop cultivation – millets, mustard etc

Animal domestication- goat, **cattle (karenj breed)**, sheep etc

15. Islands zone-

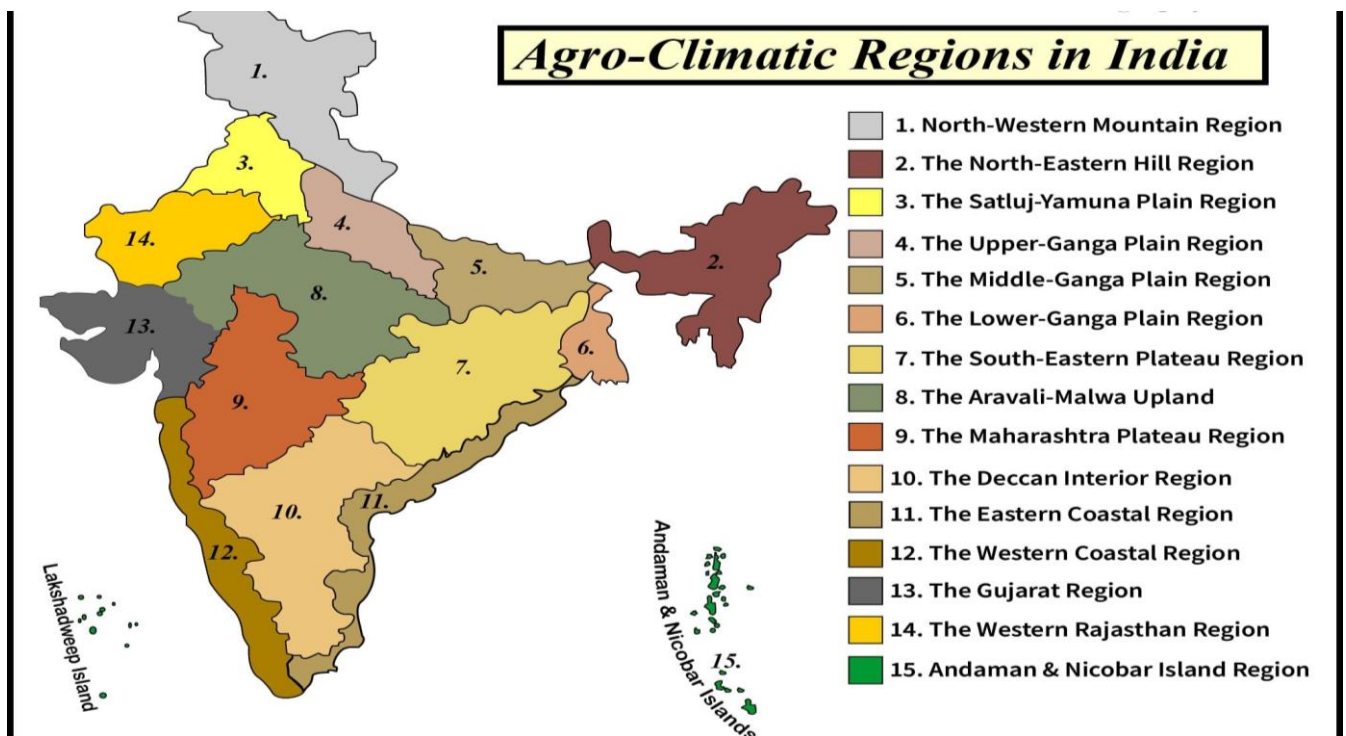
This zone covers the island **territories of Andaman and Nicobar and Lakshadweep.**

Features

- ✓ typically, equatorial **with rainfall of 3000 mm spread** over eight to nine months.
- ✓ It is **largely a forest zone** with undulated lands.

Crop cultivation- palm oil, rice spices etc agroforestry system, agrofisheery system

Animal domestication- fishery



AGROECOLOGICAL ZONES IN INDIA

Definition -Agro ecosystem are **ecological systems modified by human beings** to produce food, fiber or other agricultural products

Agro ecological zones are classified mainly on the basis of rainfall, physiography, soils & availability of irrigation water.

INDIA IS CLASSIFIED IN TO FIVE AGRO ECOSYSTEM IN INDIA: -

1. Arid
2. Rainfed
3. Irrigated
4. Coastal
5. Hill & mountain

S.NO.	AGRO ECOSYSTEM	SPECIAL FEATURES	AREA (m ha)
1.	Arid	Severe water deficit	38.7
2.	Rainfed	Occasional water deficit	67
3.	Irrigated	No water deficit	52
4	Coastal	Sandy soil, high rainfall prone to cyclones	8129 km
5.	Hill & mountain	Steep Slope & Valleys	27.5

ARID ECOSYSTEM-

In India arid ecosystem spread in **38.7 mha out of which 31.7 m ha** is under hot arid & 7 m is under cold arid ecosystem.

Features

- ✓ **Low rainfall & high potential evapotranspiration (PET)**
- ✓ Extreme **variation in diurnal & annual temperature** ex diurnal temp (day -35 to 40 ° C & at **night below 15° C** or sometimes below this.
- ✓ **Degraded soil** which contains very low organic matter.

- ✓ **Dominant vegetation-** xerophytes or ephemerals (annual plants)
- ✓ **Animal domestication-** majorly drought tolerant breeds like cattle, goat, sheep, etc.
- ✓ Major livelihood- dryland agriculture along with animal husbandry.
- ✓ Major crops- pearl millet, sesame, greengram, mustard. Etc.
- ✓ Traditional water harvesting system ex Khadin for agriculture purpose.

RAINFED ECOSYSTEM

Covers most the agricultural area in India. This area **receives annual rainfall of 500 to 1500 mm & the agricultural system** is highly dependent on the rainfall for irrigation.

Important characteristics

- ✓ **Erratic & low rainfall**
- ✓ Degraded & low fertility **soils due to flooding, water logging & excessive** leaching of nutrients
- ✓ **Harsh temp** regimes
- ✓ In rainfed **ecosystem soil acidity is the major** constrain.
- ✓ **Frequent dry spell is visible during** crop season due to break in monsoon or early withdrawal of monsoon.
- ✓ **Life saving irrigation is applied** in case of frequent dry spell.
- ✓ Soils are **mostly alfisols & vertisols**.
- ✓ **Cattle & sheep are** the major livestock in this region.

COASTAL ECOSYSTEM

Coastal areas include low lying areas **including deltas of river like ganga, region between eastern ghat & Bay of Bengal & region** between western ghat & Arabian sea

Characteristic Of This Region

- ✓ Highly fragile region **prone to cyclone, floods, tidal waves** etc. & vulnerable to municipal & industrial waste.
- ✓ **Receives high rainfall from south west monsoon** in high intensity.
- ✓ This region has higher **humidity, lower diurnal temperature** variation.
- ✓ **Coastal agroecosystem supports agriculture, aquaculture,** marine fishing, agroforestry, silviculture, Agri horticulture.

HILL & MOUNTAIN ECOSYSTEM

The major portion of this ecosystem is **located in north west Himalaya & north east Himalaya regions.**

Important characteristics

- ✓ This **region has steep slopes interspersed** with nearly level lands in valley.
- ✓ Soils **on hill & mountain slopes** are generally acidic & lateritic.
- ✓ High soil erosion.
- ✓ Soil in valley is **alluvial with loam to** sandy loam.
- ✓ This region is **characterized by high rainfall, high humidity, less solar radiation** which forms an ideal **condition for pest & disease.**
- ✓ **Majority of this region** is under plantation crops.

IRRIGATED ECOSYSTEM

This region **accounts for 37% of the agricultural land with 57% of agriculture** production. The major area is located in north Indian **i.e. Punjab, Haryana**, up, Bihar, west Bengal. In this region **the water requirement of the crops is fulfilled** by the artificial application of the water

Features

- ✓ Great diversity in **soil, water, climate resource is found in irrigated** ecosystem.
- ✓ Excess **use of surface & ground water** resource for agriculture results in water logging, soil salinity, rising water table.
- ✓ High **application of chemical fertilizer, monocropping** (rice-wheat) deteriorate the native fertility of the soil.